Lecture 10. Hypothesis Testing (Part 2)

Trend Tests, Stratified Tests, Matched Pairs, Renyi-Type Tests

Contents

1 Review: The K-Sample Test **1**

2 Tests for Trend **1**

2.1 Order-specific alternatives . . . 1

2.2 Trend test statistic . . . 2

2.3 Invariance . . . 2

3 Stratified Tests **3**

3.1 Rationale . . . 3

3.2 Set-up . . . 3

4 Matched Pairs **4**

5 Renyi-Type Tests **5**

5.1 Rationale . . . 5

5.2 Set-up and statistic . . . 5

6 Selected exercises **6**

1 Review: The K-Sample Test

Right-censored data $\{(T_{ik}, \delta_{ik})\}$; pooled death times $t_1 < \cdots < t_D$; d_j, d_{jk} deaths and Y_j, Y_{jk} at risk at t_j (pooled / group k). With a common weight W,

$$Z_k(\tau) = \sum_{j=1}^{D} W(t_j)\Big\{d_{jk} - Y_{jk}\frac{d_j}{Y_j}\Big\}, \qquad \chi^2 = (Z_1, \ldots, Z_{K-1})\,\hat{\Sigma}_{K-1}^{-1}\,(Z_1, \ldots, Z_{K-1})^T \overset{H_0}{\approx} \chi^2_{K-1},$$

with

$$\hat{\sigma}_{kk} = \sum_{j=1}^{D} W(t_j)^2 \frac{Y_{jk}}{Y_j}\Big(1 - \frac{Y_{jk}}{Y_j}\Big)\frac{Y_j - d_j}{Y_j - 1} d_j, \qquad \hat{\sigma}_{kk'} = -\sum_{j=1}^{D} W(t_j)^2 \frac{Y_{jk}Y_{jk'}}{Y_j^2}\frac{Y_j - d_j}{Y_j - 1} d_j.$$

This omnibus test is **not sensitive to a trend alternative**.

2 Tests for Trend

2.1 Order-specific alternatives

$$H_0 : h_1(t) = \cdots = h_K(t) \;\; \forall t \le \tau \qquad \text{vs.} \qquad H_A : h_1(t) \le h_2(t) \le \cdots \le h_K(t) \;\; \forall t \le \tau,$$

with at least one strict inequality; equivalently $H_A : S_1(t) \ge S_2(t) \ge \cdots \ge S_K(t)$.

2.2 Trend test statistic

Objective: increase power by a statistic sensitive to the stochastic ordering.

Definition 10.1: Trend test

For scores $a_1 < a_2 < \cdots < a_K$,

$$Z = \frac{\sum_{k=1}^{K} a_k Z_k(\tau)}{\left(\sum_{k=1}^{K}\sum_{k'=1}^{K} a_k a_{k'} \hat{\sigma}_{kk'}\right)^{1/2}} = \frac{\mathbf{a}'\mathbf{Z}}{\sqrt{\mathbf{a}'\hat{\Sigma}\mathbf{a}}},$$

with $\hat{\sigma}_{kk'}$ from the full $K \times K$ $\hat{\Sigma}$. Under H_0 each $Z_k(\tau)$ is a mean-zero martingale, so the linear combination is approximately normal with mean 0, and $Z \approx N(0,1)$. Reject H_0 in favor of H_A at level α if Z exceeds the upper α quantile of $N(0,1)$ (one-sided).

- **Choice of a_k**: heuristically, under H_A, $\mathrm{E}\{Z_1(\tau)\} \le \mathrm{E}\{Z_2(\tau)\} \le \cdots \le \mathrm{E}\{Z_K(\tau)\}$ (higher hazard $\Rightarrow$ more observed than expected deaths). Taking $a_k = k$ (standard) gives large weight to the groups with large expected values under H_A. Other choices may have better properties.
- The full $\hat{\Sigma}$ is singular, but $\mathbf{a}'\hat{\Sigma}\mathbf{a} > 0$ as long as $\mathbf{a}$ is not constant, so no group needs to be dropped.

2.3 Invariance

Theorem 10.2: Invariance under linear transformations of the scores

Let $Z_{\mathbf{a}} = \mathbf{a}'\mathbf{Z}/\sqrt{V[\mathbf{a}'\mathbf{Z}]}$ and $\mathbf{b} = c\mathbf{a} + d\mathbf{1}_K$ with constants $c > 0$, d. Then $Z_{\mathbf{b}}(\tau) = Z_{\mathbf{a}}(\tau)$.

Proof [Supplement]. Since $\sum_k Z_k(\tau) = 0$, i.e. $\mathbf{1}'\mathbf{Z} = 0$ (and hence $\hat{\Sigma}\mathbf{1} = \mathbf{0}$),

$$\mathbf{b}'\mathbf{Z} = c\,\mathbf{a}'\mathbf{Z} + d\,\mathbf{1}'\mathbf{Z} = c\,\mathbf{a}'\mathbf{Z}, \qquad V[\mathbf{b}'\mathbf{Z}] = V[c\,\mathbf{a}'\mathbf{Z}] = c^2 V[\mathbf{a}'\mathbf{Z}].$$

So $Z_{\mathbf{b}} = c\,\mathbf{a}'\mathbf{Z}/(c\sqrt{V[\mathbf{a}'\mathbf{Z}]}) = Z_{\mathbf{a}}$. (If $c < 0$ the sign flips.) ■

E.g. scores $(1,2,3,4)$ and $(0,1,2,3)$ or $(-3,-1,1,3)$ give the same test.

Example 10.3: larynx cancer by stage

90 larynx cancer patients in stages I–IV. KM curves (K&M Fig. 7.3) are ordered: stage I best, stage IV worst. Give the most severe disease the heaviest weight: $a_k = k$, $k = 1,\ldots,4$. With log-rank weights,

$$\mathbf{Z}(10.7) = (-7.5660,\ -3.0117,\ 2.9155,\ 7.6623),$$

$$\hat{\Sigma} = \begin{pmatrix} 12.0740 & -4.4516 & -6.2465 & -1.3759 \\ -4.4516 & 7.8730 & -2.7599 & -0.6614 \\ -6.2465 & -2.7599 & 9.9302 & -0.9238 \\ -1.3759 & -0.6614 & -0.9238 & 2.9612 \end{pmatrix}.$$

Then $\mathbf{a}'\mathbf{Z} = -7.566 - 6.023 + 8.747 + 30.649 = 25.806$ and $\mathbf{a}'\hat{\Sigma}\mathbf{a} = 48.152$, so

$$Z = \frac{25.806}{\sqrt{48.152}} = 3.72, \qquad Z^2 = 13.83, \qquad p < 0.0001.$$

(The value 13.83 reported in K&M is Z^2.) With other weights: Tarone–Ware $Z = 4.06$,

Gehan 4.22, Peto–Peto 4.13. All have $p < 0.0001$, strong evidence that the hazard rates are in the expected order.

Remark 10.4

Use a trend test only with strong prior information about the group ordering; otherwise it can lose power relative to the omnibus test.

3 Stratified Tests

3.1 Rationale

When comparing K groups we often need to adjust for other covariates. E.g. compare 3 vs. 6 months of chemotherapy in IBCSG adjusting for tumour stage (I–IV): the interest is the treatment, not the stage. One option is a regression model for covariate effects, but model-based adjustment is sensitive to misspecification. A conservative alternative is **stratification**.

3.2 Set-up

Definition 10.5: Stratified K-sample test

$H_0 : h_{1s}(t) = h_{2s}(t) = \cdots = h_{Ks}(t)$, $t \in [0, \tau]$, for each stratum $s = 1, \ldots, M$.

1. Using only the data in stratum s, compute $Z_{ks}(\tau)$, $k = 1, \ldots, K$, and $\hat{\Sigma}_s = \widehat{\mathrm{Cov}}[Z_{1s}(\tau), \ldots, Z_{Ks}(\tau)]$.
2. Global test: $Z_{k\cdot}(\tau) = \sum_{s=1}^{M} Z_{ks}(\tau)$ and $\hat{\Sigma}_{\cdot} = \widehat{\mathrm{Cov}}[Z_{1\cdot}, \ldots, Z_{K\cdot}] = \sum_{s=1}^{M} \hat{\Sigma}_s$ (strata are independent).
3. $\chi^2 = [Z_{1\cdot}, \ldots, Z_{(K-1)\cdot}]\, \hat{\Sigma}^{-1}_{(K-1)\times(K-1)}\, [Z_{1\cdot}, \ldots, Z_{(K-1)\cdot}]^T \overset{H_0}{\sim} \chi^2_{K-1}$.

Within each stratum, observed and expected deaths are compared only among subjects with the same stratum, so differences in the baseline hazard across strata cannot masquerade as group effects.

Example 10.6: bone marrow transplant, stratified by MTX

The three disease groups (ALL, AML low risk, AML high risk) were compared in Lecture 9 and all tests rejected H_0. A more careful comparison accounts for whether patients received the graft-versus-host prophylactic methotrexate (MTX) or not (NOMTX). Stratified log-rank ($W = 1$):

$$\text{MTX:} \quad (Z_1, Z_2, Z_3) = (1.2816,\ -2.4929,\ 1.2113),$$

$$\hat{\Sigma}_{MTX} = \begin{pmatrix} 6.23336 & -3.42915 & -2.80422 \\ -3.42915 & 5.60732 & -2.17818 \\ -2.80422 & -2.17818 & 4.98239 \end{pmatrix},$$

$$\text{NOMTX:} \quad (Z_1, Z_2, Z_3) = (-0.504,\ -11.235,\ 11.739),$$

$$\hat{\Sigma}_{NOMTX} = \begin{pmatrix} 9.6771 & -6.5535 & -3.1236 \\ -6.5535 & 13.9816 & -7.4281 \\ -3.1236 & -7.4281 & 10.5517 \end{pmatrix}.$$

Summing: $Z_{1\cdot} = 0.7776$, $Z_{2\cdot} = -13.7279$, $Z_{3\cdot} = 12.9503$ (again linearly dependent), and

$$\hat{\Sigma}_{\cdot} = \begin{pmatrix} 15.91046 & -9.98265 & -5.92782 \\ -9.98265 & 19.58892 & -9.60628 \\ -5.92782 & -9.60628 & 15.53409 \end{pmatrix}.$$

Using the first two:

$$\chi^2 = (0.7776,\ -13.7279) \begin{pmatrix} 15.9105 & -9.9827 \\ -9.9827 & 19.5889 \end{pmatrix}^{-1} \begin{pmatrix} 0.7776 \\ -13.7279 \end{pmatrix} = 13.19 \quad (2 \text{ df},\ p \approx 0.001).$$

Remark 10.7: Remarks

Stratified tests have good power against alternatives in the *same direction* in every stratum. Otherwise, Z from one stratum can be cancelled by Z from another, the overall statistic may be near 0, and the test has very low power. In that case, run separate K-sample tests in each stratum.

4 Matched Pairs

Another use of stratification: each matched pair is a stratum. Paired times (T_{1i}, T_{2i}) and indicators $(\delta_{1i}, \delta_{2i})$, $i = 1, \ldots, M$; test $H_0 : h_{1i}(t) = h_{2i}(t)$. With $K = 2$, in each pair

$$Z_{1i}(\tau) = \sum_j W(t_j)\Big\{d_{j1} - Y_{j1}\frac{d_j}{Y_j}\Big\}, \qquad \sigma_{11i} = \sum_j W(t_j)^2 \frac{Y_{j1}}{Y_j}\Big(1 - \frac{Y_{j1}}{Y_j}\Big)\frac{Y_j - d_j}{Y_j - 1} d_j.$$

Derivation: within a single pair

Before the first time in the pair both are at risk: $Y_j = 2$, $Y_{j1} = 1$, so $Y_{j1}/Y_j = 1/2$.

1. $T_{1i} = T_{2i}$: $d_j = \delta_{1i} + \delta_{2i}$,

$$Z_{1i} = W(T_{1i})\Big\{\delta_{1i} - \frac{\delta_{1i} + \delta_{2i}}{2}\Big\} = W(T_{1i})\frac{\delta_{1i} - \delta_{2i}}{2},$$

$$\sigma_{11i} = W^2(T_{1i})\tfrac{1}{2}(1 - \tfrac{1}{2})\frac{2 - \delta_{1i} - \delta_{2i}}{2 - 1}(\delta_{1i} + \delta_{2i})$$

$$= \frac{W^2(T_{1i})}{4}(2 - \delta_{1i} - \delta_{2i})(\delta_{1i} + \delta_{2i}).$$

2. $T_{1i} < T_{2i}$: only the first time can be an event with both at risk,

$$Z_{1i} = W(T_{1i})\Big\{\delta_{1i} - \frac{\delta_{1i}}{2}\Big\} = W(T_{1i})\frac{\delta_{1i}}{2}, \qquad \sigma_{11i} = \frac{W^2(T_{1i})}{4}(2 - \delta_{1i})\delta_{1i}.$$

3. $T_{1i} > T_{2i}$: $Z_{1i} = -W(T_{2i})\dfrac{\delta_{2i}}{2}$, $\quad \sigma_{11i} = \dfrac{W^2(T_{2i})}{4}(2 - \delta_{2i})\delta_{2i}$.

(After the first time, one of the pair is gone, so $Y_{j1}/Y_j \in \{0, 1\}$ and later terms contribute nothing.) For 0/1 indicators $(2 - \delta)\delta = \delta$, so:

$$Z_{1i} = \begin{cases} W(T_{1i})/2, & T_{1i} < T_{2i}, \delta_{1i} = 1 \text{ or } T_{1i} = T_{2i}, \delta_{1i} = 1, \delta_{2i} = 0, \\ -W(T_{2i})/2, & T_{2i} < T_{1i}, \delta_{2i} = 1 \text{ or } T_{1i} = T_{2i}, \delta_{2i} = 1, \delta_{1i} = 0, \\ 0, & \text{otherwise;} \end{cases}$$

$$\hat{\sigma}_{11i} = \begin{cases} W(T_{1i})^2/4, & \text{first case,} \\ W(T_{2i})^2/4, & \text{second case,} \\ 0, & \text{otherwise.} \end{cases}$$

Theorem 10.8: Paired log-rank (sign-type) test

If W does not depend on time (e.g. log-rank, $W \equiv 1$), with $D_1 = \#$ pairs in which member 1 has the event first and $D_2 = \#$ pairs in which member 2 does,

$$Z_{1\cdot} = \frac{W}{2}(D_1 - D_2), \qquad \hat{\sigma}_{11\cdot} = \sum_{i=1}^{M} \hat{\sigma}_{11i} = \frac{W^2}{4}(D_1 + D_2), \qquad \frac{Z_{1\cdot}}{\sqrt{\hat{\sigma}_{11\cdot}}} = \frac{D_1 - D_2}{\sqrt{D_1 + D_2}} \overset{H_0}{\approx} N(0,1).$$

This is the sign test (McNemar-type): under H_0 each informative pair is equally likely to go either way.

Example 10.9: 6-MP vs. placebo matched pairs

$D_{\text{Placebo}} = 18$, $D_{\text{6-MP}} = 3$, so $Z = (18-3)/\sqrt{18+3} = 3.27$ and $p = 2P[Z > 3.27] = 0.001$: sufficient evidence that the relapse rates differ.

5 Renyi-Type Tests

5.1 Rationale

If the hazards of the two groups **cross**, the usual two-sample tests may have low power: early differences cancel late differences of opposite sign. Renyi-type tests are built on the *absolute values of cumulative differences*; a Renyi statistic is the analogue for censored data of the Kolmogorov–Smirnov statistic.

5.2 Set-up and statistic

Test $H_0 : h_1(t) = h_2(t)$, $t \le \tau$, vs. $H_A : h_1(t) \neq h_2(t)$ for some $t \le \tau$. Two samples of sizes n_1, n_2; pooled distinct failure times $t_1 < \cdots < t_D$; d_{j1}, d_{j2} and Y_{j1}, Y_{j2} with $d_j = d_{j1} + d_{j2}$, $Y_j = Y_{j1} + Y_{j2}$.

Definition 10.10: Renyi statistic

Cumulative weighted difference up to t_i:

$$Z(t_i) = \sum_{t_j \le t_i} W(t_j)\Big[d_{j1} - Y_{j1}\frac{d_j}{Y_j}\Big], \qquad i = 1, \ldots, D,$$

with standard error at τ (largest t_j with $Y_{j1}, Y_{j2} > 0$)

$$\sigma^2(\tau) = \sum_{t_j \le \tau} W^2(t_j)\frac{Y_{j1}}{Y_j}\frac{Y_{j2}}{Y_j}\frac{Y_j - d_j}{Y_j - 1}d_j.$$

The two-sided test statistic is

$$Q = \frac{\sup\{|Z(t)|,\ t \le \tau\}}{\sigma(\tau)}.$$

Theorem 10.11: Null distribution

Under H_0, Q is approximately distributed as $\sup\{|B(x)|,\ 0 \le x \le 1\}$, B a standard Brownian motion. Critical values are in K&M Table C.5.

Remark 10.12: Why Brownian motion

$Z(t)$ is a martingale in t with variance process $\sigma^2(t)$; rescaling time by $x = \sigma^2(t)/\sigma^2(\tau) \in [0,1]$, $Z/\sigma(\tau)$ behaves like $B(x)$. The supremum of $|Z|$ catches a large excursion even if $Z(\tau)$ returns near 0.

Comparison with the usual test. The usual statistic is $Z(\tau)/\sqrt{\mathrm{Var}[Z(\tau)]}$ (or its square). When hazards cross, early positive differences are cancelled by later negative ones, and the log-rank test may lack power; the supremum test based on $|Z|$ has greater power for such differences.

Example 10.13: gastric cancer: chemotherapy vs. chemotherapy + radiotherapy

The KM curves cross (K&M Fig. 7.5): chemo + radiation is worse early but better later. Using the log-rank version of the Renyi statistic, $|Z(t_j)|$ (K&M Fig. 7.4) peaks at $t_j = 315$ with value 9.8; $\sigma(\tau = 2363) = 4.46$, so

$$Q = 9.8/4.46 = 2.2, \qquad 0.05 < p < 0.06 \text{ (K\&M Table C.5)},$$

close to rejecting H_0. The usual log-rank test gives $Z(2363) = -2.15$, statistic $-2.15/4.46 = -0.482$, $p = 2P(Z < -0.482) = 0.63$: it misses the crossing difference entirely.

6 Selected exercises

Motivation. Two claims of this lecture deserve a proof: when the trend-test variance $\mathbf{a}'\hat{\Sigma}\mathbf{a}$ is really positive, and why the paired log-rank test is an exact sign test under the null hypothesis.

Exercise 10.1 (When is the trend test defined?)

Write $\pi_{jk} = Y_{jk}/Y_j$ and $w_j = W(t_j)^2 d_j(Y_j - d_j)/(Y_j - 1)$, with $w_j = 0$ if $Y_j = 1$.

(a) Show that $\mathbf{a}'\hat{\Sigma}\mathbf{a} = \sum_j w_j \Big\{ \sum_k \pi_{jk} a_k^2 - \Big(\sum_k \pi_{jk} a_k \Big)^2 \Big\} \ge 0$.

(b) Show that $\mathbf{a}'\hat{\Sigma}\mathbf{a} > 0$ if and only if at some death time with $w_j > 0$ the scores a_k are not all equal over the groups with $Y_{jk} > 0$. In particular, for $a_1 < \cdots < a_K$ it suffices that at one death time with $W(t_j) \neq 0$ and $d_j < Y_j$ at least two groups are at risk.

(c) For $K = 2$ and $a_1 < a_2$, show that the trend statistic equals $Z_2(\tau)/\sqrt{\hat{\sigma}_{22}} = -Z_1(\tau)/\sqrt{\hat{\sigma}_{11}}$.

Solution [Supplement].

(a) By the formulas for $\hat{\sigma}_{kk'}$, the contribution of t_j to $\hat{\Sigma}$ is $w_j\{\mathrm{diag}(\boldsymbol{\pi}_j) - \boldsymbol{\pi}_j\boldsymbol{\pi}_j'\}$. So $\mathbf{a}'\hat{\Sigma}\mathbf{a} = \sum_j w_j\{\sum_k \pi_{jk}a_k^2 - (\sum_k \pi_{jk}a_k)^2\}$. Since $\pi_{jk} \ge 0$ and $\sum_k \pi_{jk} = 1$, the brace is the variance of a_κ when the group label κ is drawn with probabilities $\boldsymbol{\pi}_j$, hence ≥ 0; also $w_j \ge 0$.

(b) A variance vanishes iff the variable is a.s. constant, i.e. iff a_k is constant on $\{k : \pi_{jk} > 0\} = \{k : Y_{jk} > 0\}$. A sum of nonnegative terms is positive iff some term is, which proves the equivalence. For strictly increasing scores, two groups at risk make the a_k unequal, and $W(t_j) \neq 0$, $1 \le d_j < Y_j$ give $w_j > 0$.

(c) Since $Z_1 = -Z_2$ and, by Exercise 9.3(a), $\hat{\sigma}_{11} = -\hat{\sigma}_{12} = \hat{\sigma}_{22}$,

$$\mathbf{a}'\mathbf{Z} = (a_2 - a_1)Z_2, \qquad \mathbf{a}'\hat{\Sigma}\mathbf{a} = (a_2 - a_1)^2\hat{\sigma}_{22}.$$

Dividing, and using $a_2 - a_1 > 0$, gives $Z_2/\sqrt{\hat{\sigma}_{22}} = -Z_1/\sqrt{\hat{\sigma}_{11}}$: for two groups the trend test is the one-sided two-sample test of $h_1 \le h_2$. ■

Exercise 10.2 (The paired log-rank test as an exact sign test)

In pair i ($i = 1, \dots, M$, pairs independent) let X_{1i}, X_{2i} be i.i.d. continuous and both censored at a common time C_i independent of (X_{1i}, X_{2i}). Let D_1 (D_2) be the number of pairs in which member 1 (member 2) has an observed event first, and $m = D_1 + D_2$.
(a) Show that, conditionally on m, $D_1 \sim \mathrm{Bin}(m, 1/2)$.
(b) Show that $\mathrm{E}(D_1 - D_2 \mid m) = 0$ and $\mathrm{Var}(D_1 - D_2 \mid m) = m$, matching the paired log-rank statistic.
(c) Compute the exact two-sided p-value for the 6-MP pairs ($D_{\text{Placebo}} = 18$, $D_{\text{6-MP}} = 3$).

Solution [Supplement].

(a)

Step 1. Member 1 has an observed event first iff $A_{1i} = \{X_{1i} < X_{2i},\ X_{1i} \le C_i\}$, and similarly A_{2i} (ties have probability 0). These events are disjoint.

Step 2. Swapping X_{1i} and X_{2i} leaves the joint law of (X_{1i}, X_{2i}, C_i) unchanged and maps A_{1i} onto A_{2i}, so $P(A_{1i}) = P(A_{2i})$.

Step 3. Let $\xi_i = 1, 2, 0$ according as A_{1i}, A_{2i}, or neither occurs. The ξ_i are independent, and $P(\xi_i = 1 \mid \xi_i \neq 0) = 1/2$ by Step 2. Conditionally on the set $J = \{i : \xi_i \neq 0\}$, the ξ_i, $i \in J$, are independent fair choices between 1 and 2, so $D_1 \mid J \sim \mathrm{Bin}(|J|, 1/2)$. This depends on J only through $|J| = m$, so the same holds given m.

(b) $D_1 - D_2 = 2D_1 - m$, so $\mathrm{E}(D_1 - D_2 \mid m) = 2 \cdot \frac{m}{2} - m = 0$ and $\mathrm{Var}(D_1 - D_2 \mid m) = 4 \cdot \frac{m}{4} = m$. With $W \equiv 1$ the paired statistic is $(D_1 - D_2)/\sqrt{D_1 + D_2}$, exactly the standardized difference.

(c) With $m = 21$ and $D_1 = 18$,

$$p = 2P\{\mathrm{Bin}(21, \tfrac{1}{2}) \ge 18\} = 2 \cdot \frac{\binom{21}{18} + \binom{21}{19} + \binom{21}{20} + \binom{21}{21}}{2^{21}} = \frac{2(1330 + 210 + 21 + 1)}{2\,097\,152} \approx 0.0015,$$

in agreement with the normal approximation $p \approx 0.001$. ■